

Advanced Topics in Deep Learning

Knowledge Distillation with Ternary-Weight Quantization-Aware Training

Comprehensive Final Report

Teacher: CIFAR-adapted ResNet34 FP32

Student: CIFAR-adapted ResNet18 with strict ternary Conv/FC deployment

Dataset: CIFAR-10, fixed 45,000/5,000 train/validation split plus one final 10,000-image official test evaluation

Final report generation: 2026-09-11 16:29 IST

Executive result. The frozen vanilla ternary-KD control achieved 94.87% +/- 0.09% on the official unseen CIFAR-10 test set across three independently trained, validation-selected checkpoints. This exceeds the ternary no-KD baseline by 0.18 percentage points on test accuracy while retaining strict ternary Conv/FC deployment.

Integrity statement

All architecture and hyperparameter choices were made using the fixed validation split. The final official test report was generated only after the teacher, baselines, final vanilla-KD control, and exploratory QTRD checkpoint were frozen. No test-based model selection, tuning, restart, or overwrite was permitted. The official evaluator records independently recomputed validation accuracy, test accuracy, class confusion matrices, and ternary-verification diagnostics for every tested checkpoint.

1. Problem, protocol, and reproducibility

The project trains a CIFAR-adapted ResNet34 teacher from scratch and transfers knowledge to ResNet18 students. The production data pipeline fixes a class-balanced 45,000/5,000 split from CIFAR-10 training examples (seed 42), with exact train and validation index hashes stored by the pipeline. Training uses random crop plus horizontal flip; validation and official test use deterministic normalization. CUDA execution, package locks, per-run configuration, seeds, checkpoints, histories, and reload checks are preserved.

The official test set was inaccessible during all development. The one-time final evaluator was explicitly unlocked after the validation-selected checkpoints were frozen. It produces *results/final_official_test_evaluation.json* immutably; a second invocation refuses to overwrite the report.

2. Model and ternary-QAT design

The teacher is a CIFAR ResNet34 with a 3x3, stride-1 stem, no ImageNet max-pool, and a 10-class classifier. The student is the matching CIFAR ResNet18. In the compliant student, every convolution and the final fully connected layer deploys ternary weights. BatchNorm and biases remain FP32. For latent weight W , the per-output-channel threshold is $\Delta = 0.7 \text{ mean}(|W|)$; the code $Q(W)$ is in $\{-1, 0, +1\}$; and the deployed weight is $\hat{W} = \alpha Q(W)$, where α is the mean active absolute latent weight. Latent FP32 weights are optimized, while $\hat{W}$ is used in every forward pass.

Backpropagation uses a clipped straight-through estimator. This is genuine QAT rather than post-hoc ternarization: quantization is active in the forward pass throughout optimization. The saved ternary checkpoints are rebuilt and verified layer by layer: deployed values match the quantizer, every Conv/FC is covered, and latent parameters contain more than three values.

3. KD formulation and training

The final control uses vanilla KD: $(1-\lambda) \text{CE}(y, s) + \lambda T^2 \text{KL}(\text{softmax}(t/T) \parallel \text{softmax}(s/T))$, with $T=2$ and $\lambda=0.9$. The ResNet34 teacher is loaded from its frozen checkpoint, switched to eval mode, excluded from the optimizer, and evaluated under no-grad. Student training uses SGD with momentum 0.9, Nesterov, warmup plus cosine learning rate schedule, label smoothing 0.1, and the fixed validation protocol. The final control was trained for 200 epochs with seeds 42, 43, and 44.

4. Methods explored and decisions

Method	Evidence	Validation result	Decision
FP32 ResNet34 teacher	200 epochs, one frozen teacher	95.54%	reference
FP32 ResNet18 baseline	200 epochs, 3 seeds	95.34% +/- 0.14%	baseline
Ternary ResNet18, no KD	200 epochs, 3 seeds	95.21% +/- 0.08%	quantization reference
Vanilla ternary KD	T=2, lambda=0.9, 200 epochs, 3 seeds	95.12% +/- 0.15%	frozen final KD control
DKD	3 seeds	94.88% +/- 0.16%	rejected below control
DIST	one-seed screen	94.94%	screening only
DIST continuation	T=1, lambda=0.5, one seed	95.02%	provisional below control
QFD / attention transfer / RKD	one seed each	95.00% / 95.02% / 95.12%	provisional screens
QTRD	one seed, 50 epochs	95.06%	exploratory; below control

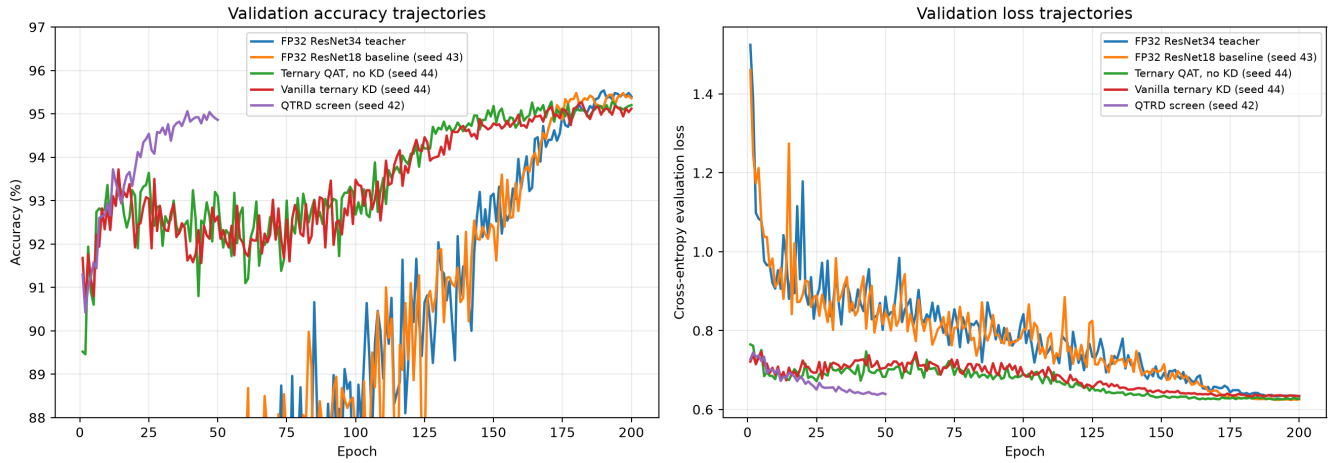
Automated research infrastructure was used to preserve staged configurations, append-only artifacts, diagnostics, and a research ledger. Open-ended AutoML/HPO was intentionally paused rather than allowed to conduct opaque or test-driven search. The published final selection is the pre-test, three-seed vanilla-KD aggregate; advanced one-seed methods are clearly labelled exploratory and are not promoted to final claims.

5. Final official unseen-test evaluation

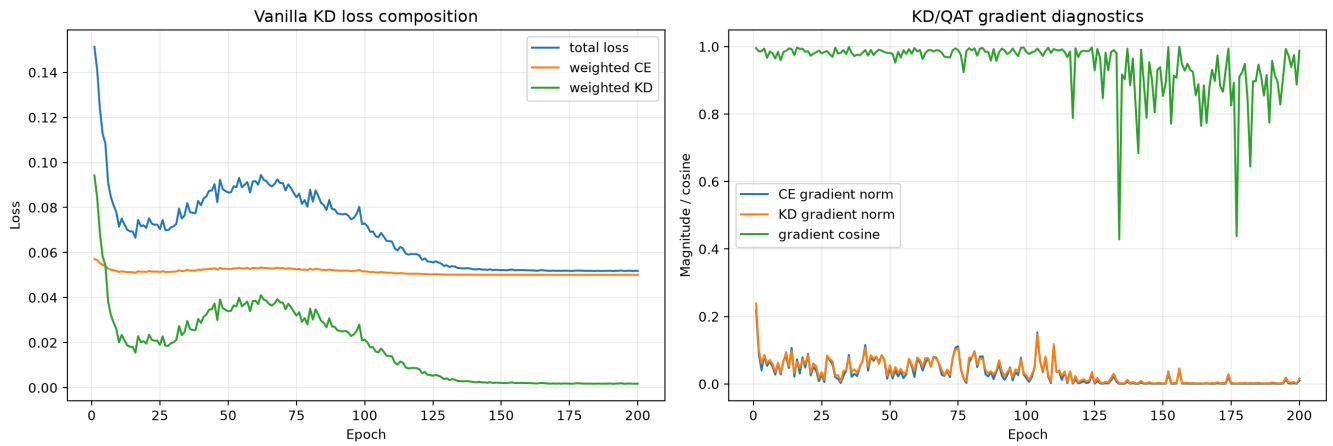
Model	Validation mean	Official test mean	n	Interpretation
FP32 ResNet18 baseline	95.34%	95.01% +/- 0.08%	3	full-precision baseline
FP32 ResNet34 teacher	95.54%	95.19% +/- 0.00%	1	teacher reference
QTRD exploratory screen	95.06%	94.83% +/- 0.00%	1	screen only; not selected
Ternary ResNet18 (no KD)	95.21%	94.69% +/- 0.04%	3	quantization cost reference
Vanilla ternary KD (final control)	95.12%	94.87% +/- 0.09%	3	selected before test access

The final vanilla-KD control improves official test accuracy over the no-KD ternary baseline by 0.18 percentage points (94.87% versus 94.69%). It remains 0.14 percentage points below the FP32 ResNet18 baseline and 0.32 percentage points below the ResNet34 teacher, quantifying the accuracy/compression trade-off honestly.

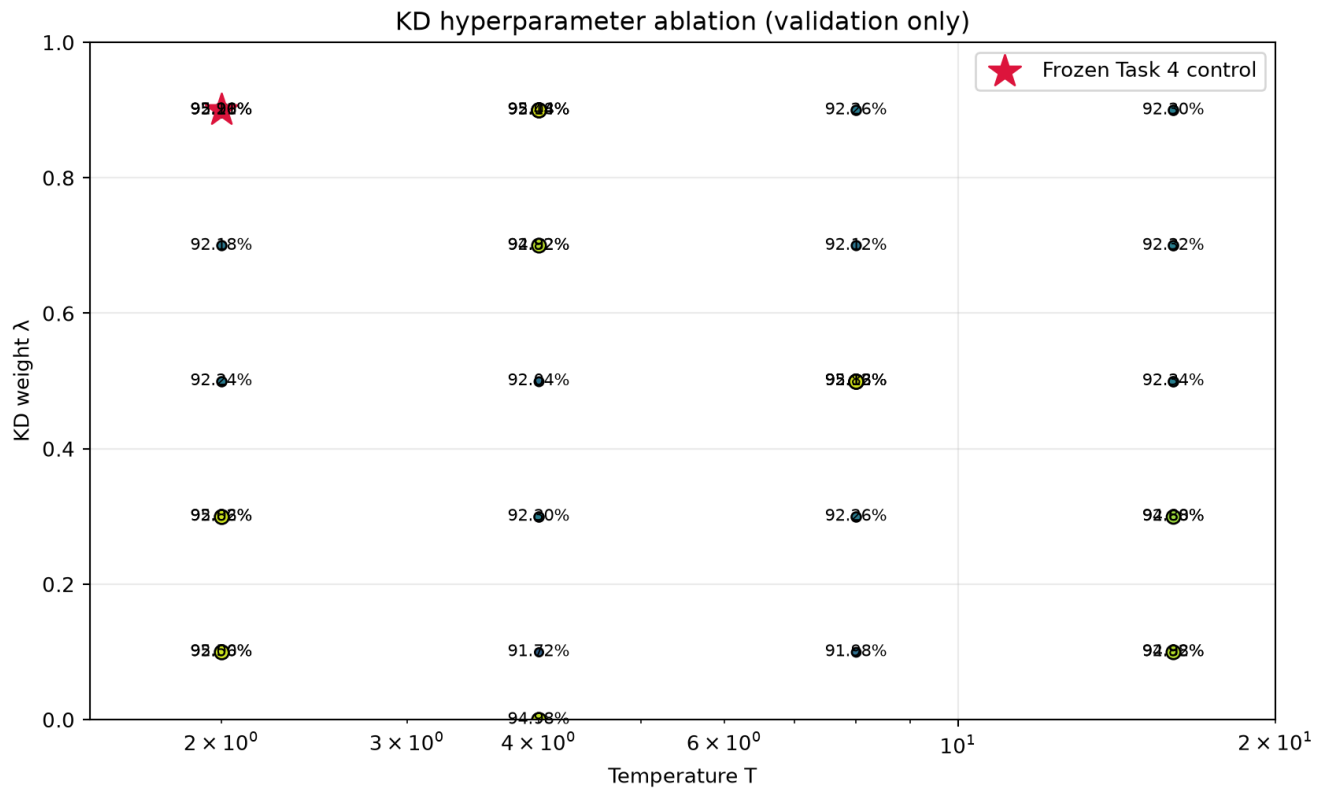
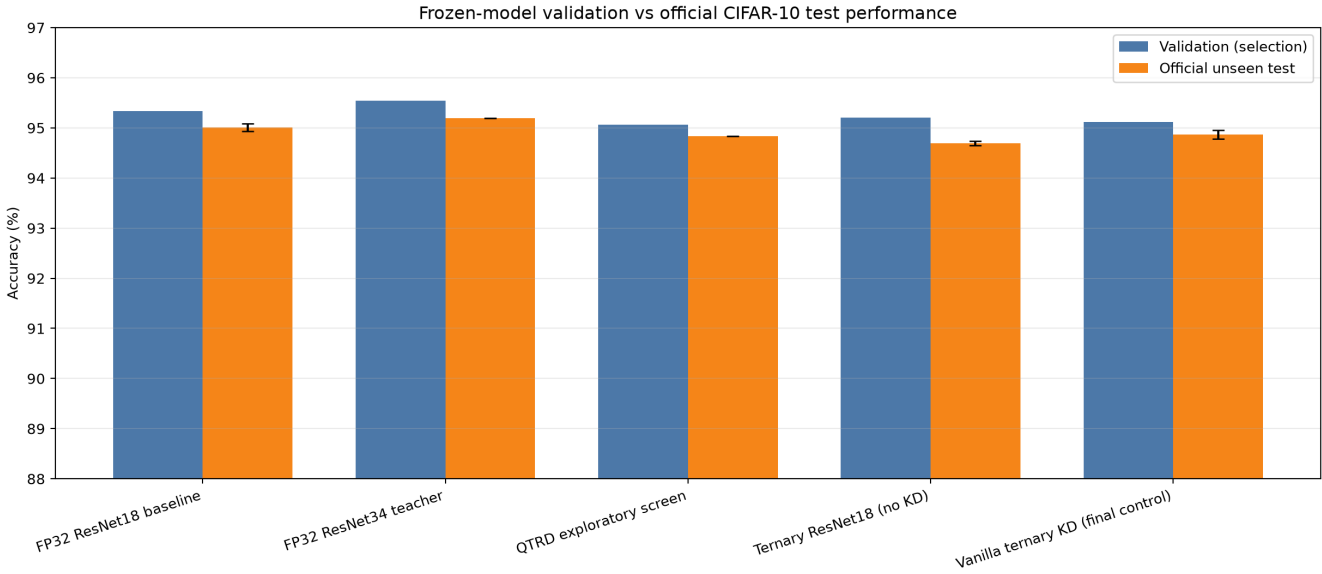
6. Training curves and KD stability



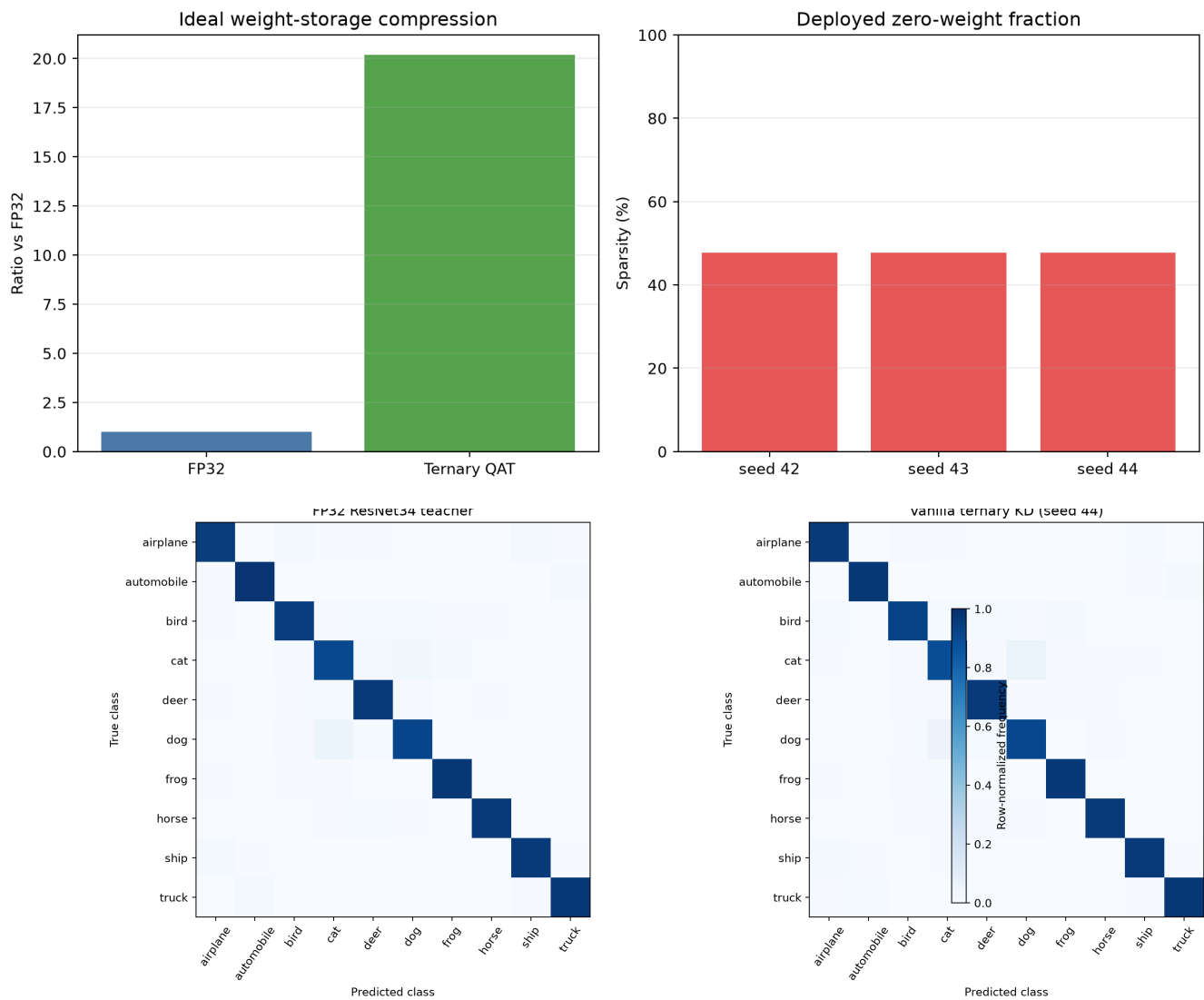
The trajectories compare the teacher, FP32 ResNet18, pre-KD ternary QAT, post-KD vanilla ternary QAT, and QTRD screen using real preserved histories. The shorter QTRD curve is deliberately not visually or statistically equated with the 200-epoch, three-seed control.



7. Accuracy, ablation, compression, and error analysis



8. Compression and classwise behavior



9. Checkpoints, storage, and execution time

Artifact	Parameters	On-disk checkpoint	Interpretation
FP32 ResNet34 teacher	21,282,122	162.54 MiB	FP32 deployed
FP32 ResNet18 baseline	11,173,962	85.35 MiB	FP32 deployed
Ternary ResNet18 no KD	11,173,962	85.35 MiB	latent FP32 training checkpoint
Vanilla ternary KD control	11,173,962	85.35 MiB	latent FP32 training checkpoint
QTRD exploratory	11,173,962	85.35 MiB	latent FP32 training checkpoint

The approximately 85 MiB ternary checkpoint files intentionally retain latent FP32 training weights, optimizer-independent state, and metadata so they can be resumed and verified. They are not deployment packages. The deployed Conv/FC representation requires $\log_2(3)$ bits per weight plus scale metadata, giving an ideal weight-storage ratio of 20.19x versus FP32 before packing, scale overhead, and runtime/kernel effects. The final-control ternary verification reports approximately 47% zero values; no hardware speedup is claimed.

Phase	Recorded cumulative GPU training time
Teacher (one 200-epoch run)	0h 55m 17s
FP32 ResNet18 baseline (3 seeds)	1h 26m 30s
Ternary no-KD QAT baseline (3 seeds)	1h 50m 57s
Vanilla ternary KD final control (3 seeds)	2h 11m 58s
Advanced KD/QAT screens and confirmations	3h 25m 10s
Recorded cumulative total	9h 49m 52s

Times are sums of per-run saved elapsed times and represent active training duration; they exclude dataset preparation, notebook startup, report generation, and final inference-only evaluation. This makes the stated total auditable rather than a speculative wall-clock estimate.

10. Limitations and critical discussion

The ternary result is a storage-oriented algorithmic compression result. Practical inference acceleration requires ternary-aware kernels, efficient packing, memory-transfer analysis, and deployment benchmarking; these were not claimed or measured. The model uses a fixed split and multiple seed checks, but exact deterministic equivalence is not claimed for every CUDA operation. Advanced transfer losses beyond vanilla KD were intentionally constrained to one-seed screens unless evidence justified further expenditure; they should not be interpreted as confirmatory comparisons. Representation-similarity metrics were not logged in the completed early runs and are therefore not fabricated.

11. Required references

- [1] K. He, X. Zhang, S. Ren, and J. Sun. Deep Residual Learning for Image Recognition. CVPR, 2016.
- [2] G. Hinton, O. Vinyals, and J. Dean. Distilling the Knowledge in a Neural Network. NeurIPS Deep Learning Workshop, 2015.
- [3] F. Li, B. Zhang, and B. Liu. Ternary Weight Networks. arXiv:1605.04711, 2016.
- [4] C. Zhu, S. Han, H. Mao, and W. J. Dally. Trained Ternary Quantization. ICLR, 2017.
- [5] Y. Bengio, N. Leonard, and A. Courville. Estimating or Propagating Gradients Through Stochastic Neurons. arXiv:1308.3432, 2013.
- [6] P. Yin, S. Lyu, R. Zhang, S. Osher, Y. Qi, and J. Xin. Understanding Straight-Through Estimator in Training Activation Quantized Neural Nets. ICLR, 2019.

Authoritative artifacts

Official results: results/final_official_test_evaluation.json. Visual notebook: ATDL_final_official_evaluation_visual_report.ipynb. Final selected checkpoint: experiments/task4/checkpoints/task4_b3_final_t2_lam09_remaining_rerun1/resnet18_ternary_kd_seed44.pth. Ternary verification tool: src/evaluation/verify_ternary.py. This report supersedes the earlier validation-only final report for submission-facing claims.